

Homework 1

Submission date: 1st April, 2019

1. In terms of the Lagrange Formula $(\mathbf{a} \times \mathbf{b}) \cdot (\mathbf{c} \times \mathbf{d}) = (\mathbf{a} \cdot \mathbf{c})(\mathbf{b} \cdot \mathbf{d}) - (\mathbf{a} \cdot \mathbf{d})(\mathbf{b} \cdot \mathbf{c})$

Please prove the following equation is true. ($\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ are all tensors)

$$e_{ksp}e_{ipj} = \delta_{is}\delta_{jk} - \delta_{ik}\delta_{js},$$

2. Please prove that velocity $\mathbf{v}$ and accelerated velocity $\mathbf{a}$ are vectors.
3. Please prove that permutation symbol e_{ijk} is a three-order tensor.
4. Please prove that strain $\boldsymbol{\varepsilon}$ is a second-order tensor.

5. Tensor T is known, $T_{ij} = \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 2 \\ 0 & 2 & 3 \end{bmatrix}$, please solve the following

questions: (1) T_{ii} , (2) $T_{ij}T_{ij}$